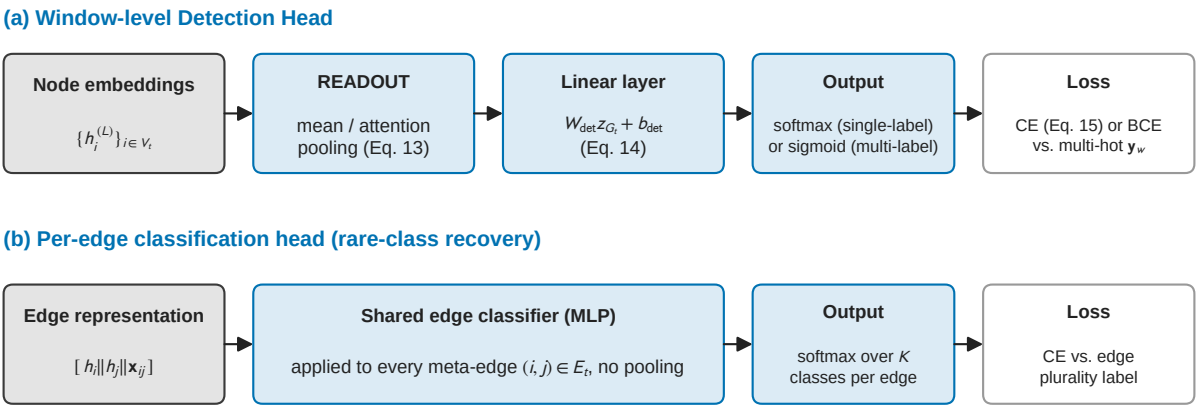
(a) Window-level Detection Head
Node embeddings
$\{h_i^{(L)}\}_{i \in V_t}$
READOUT
mean / attention pooling (Eq. 13)
Linear layer
$W_{\text{det}} z_{G_t} + b_{\text{det}}$ (Eq. 14)
Output
softmax (single-label) or sigmoid (multi-label)
Loss
CE (Eq. 15) or BCE vs. multi-hot $\mathbf{y}_w$
(b) Per-edge classification head (rare-class recovery)
Edge representation
$[\, h_i \| h_j \| \mathbf{x}_{ij} \,]$
Shared edge classifier (MLP)
applied to every meta-edge $(i, j) \in E_t$, no pooling
Output
softmax over K classes per edge
Loss
CE vs. edge plurality label